Response to Reviewers

Manuscript PONE-D-26-28965

Evaluation of validation strategies and transferability of EEG attention decoding: a comparative analysis on open datasets

[TODO — working state of this letter; this banner is not part of the submission.] Every response below is in one of three states. A response with no marker is backed both by the code in `python_project/` and by text that is actually present in `manuscript_clean/manuscript.tex`. A response followed by a **[TODO]** paragraph is only partly backed: the implementation exists, but the manuscript does not yet say so, or the numbers it refers to have not been computed. A response consisting only of **[TODO]** is not answered at all yet. The manuscript is mid-rewrite — its Abstract, Background, Models and Results sections are cleared placeholders — the full sweep in `python_project/results/` is still running, and no analysis has been produced from it. Until that changes, no response here may cite a number, a table, a figure or a manuscript passage. All markers must be gone before the letter is uploaded.

Dear Editor and Reviewers,

We thank the Editor and the Reviewers for their careful evaluation of our manuscript and for their constructive comments. We have carefully considered all comments and are preparing a revised version of the manuscript that addresses the issues raised during peer review.

The major revisions will include:

- **[TODO — major revision 1: redefinition of the target construct.]** The direction is settled — the study claims discrimination between operationally defined experimental conditions rather than decoding of attention as a psychological construct — but the manuscript does not yet carry it: Table 1 still names its classes "High Attention" and "Low Attention", and no label-mapping sensitivity analysis exists in the code. Only the exclusion of drowsy epochs from the primary analysis is in place. Rewrite this bullet together with the responses to MR1 and Comments 3.1, 3.2 and 4.5.
- *[Major revision 2 — to be completed after the corresponding manuscript changes are finalized.]*
- *[Major revision 3 — to be completed after the corresponding manuscript changes are finalized.]*
- *[Additional major revision, if needed.]*

For clarity, our response is organized into four sections:

1. **Journal Requirements**
2. **Reviewer #1**
3. **Reviewer #2**
4. **Reviewer #3**

Reviewer and journal comments are reproduced below and followed by our point-by-point responses. Where a single original paragraph contains several independent criticisms or requests, it is divided into separate response points so that each issue can be addressed explicitly. Changes to the manuscript are identified with the corresponding page and line numbers wherever applicable.

Yours sincerely,

[TODO: author list]

Journal Requirements

The seven requirements listed in the decision letter are addressed below.

Comment 1

Please ensure that your manuscript meets PLOS ONE's style requirements, including those for file naming. The PLOS ONE style templates can be found at https://journals.plos.org/plosone/s/file?id=wjVg/PLOSOne_formatting_sample_main_body.pdf and https://journals.plos.org/plosone/s/file?id=ba62/PLOSOne_formatting_sample_title_authors_affiliations.pdf

Response:

[TODO]

Comment 2

Please note that PLOS One has specific guidelines on code sharing for submissions in which author-generated code underpins the findings in the manuscript. In these cases, we expect all author-generated code to be made available without restrictions upon publication of the work. Please review our guidelines at https://journals.plos.org/plosone/s/materials-and-software-sharing#loc-sharing-code and ensure that your code is shared in a way that follows best practice and facilitates reproducibility and reuse.

Response:

[TODO]

Comment 3

We note that the grant information you provided in the 'Funding Information' and 'Financial Disclosure' sections do not match. When you resubmit, please ensure that you provide the correct grant numbers for the awards you received for your study in the 'Funding Information' section.

Response:

[TODO]

Comment 4

Thank you for stating the following financial disclosure:
"The author(s) declared that financial support was received for this work and/or its publication. This research was funded by the Committee of Science of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant no. BR27198099)."
Please state what role the funders took in the study. If the funders had no role, please state: "The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript."
If this statement is not correct you must amend it as needed.
Please include this amended Role of Funder statement in your cover letter; we will change the online submission form on your behalf.

Response:

[TODO]

Comment 5

Please note that your Data Availability Statement is currently missing the repository name and/or the DOI/accession number of each dataset OR a direct link to access each database. If your manuscript is accepted for publication, you will be asked to provide these details on a very short timeline. We therefore suggest that you provide this information now, though we will not hold up the peer review process if you are unable.

Response:

[TODO]

Comment 6

We notice that your supplementary figures are uploaded with the file type 'Figure'. Please amend the file type to 'Supporting Information'. Please ensure that each Supporting Information file has a legend listed in the manuscript after the references list.

Response:

[TODO]

Comment 7

If the reviewer comments include a recommendation to cite specific previously published works, please review and evaluate these publications to determine whether they are relevant and should be cited. There is no requirement to cite these works unless the editor has indicated otherwise.

Response:

[TODO]

Reviewer #1

Comment 1

MR1. The binary attention labels have uncertain construct validity. Dataset B was collected for stress analysis rather than attention measurement, while Dataset A uses instructed states presented in a fixed temporal sequence. Drowsiness also involves eye closure. Consequently, the classes confound attention with stress, arousal, visual input, ocular activity and time-on-task.

Response:

[TODO — this response must be written again from scratch; the previous draft described a manuscript that does not exist.] It claimed a renamed Table 1 with operationally defined classes, a paragraph after Table 1 stating what the labels do not denote, a Limitations subsection treating the Reviewer's five confounds one by one, a Mühl et al. (2014) framing of stress as an affective context, and five label-mapping sensitivity arms. None of these are in `manuscript.tex`: Table 1 is unchanged ("Unified binary attention label mapping", classes "High Attention" and "Low Attention"), there is no Limitations subsection at all although two passages of Methodology already refer the reader to one, `muhl2014workload` sits in `refs.bib` uncited, and neither the configuration nor the sweep contains any label-mapping arm beyond `dataset.exclude_conditions`.

[TODO — what is true today and may be reused.] Drowsy epochs are excluded from the primary analysis, and the exclusion happens at the head of the preparation pipeline, before channel selection, filtering, ICA and windowing, so that no drowsy sample reaches the ICA decomposition or the feature scaler. This holds in the code and is stated in the Data preparation subsection. The two factual observations about the datasets — that the authors of Dataset A report that no independent objective control could be introduced, and that the public release of Dataset B omits the participants' responses — are also sound and can be carried over.

[TODO — what the answer has to argue.] The construct problem is not solved by renaming the classes. The answer must concede that *attention* is an umbrella term with no single operational definition, reframe the target accordingly, and present the label mapping as an explicit operational choice whose consequences are measured rather than assumed; see `thinking_about_introduction_section/` in the repository root. It can only be written after the corresponding paragraphs exist in the Introduction and Materials sections of the manuscript.

Comment 2

MR2. The principal within-dataset baseline is affected by substantial information leakage. One-second windows separated by only 0.125 seconds share 87.5% of their samples, yet temporally adjacent windows are distributed across stratified folds. Acknowledging this issue as a limitation does not make the reported baseline scientifically valid.

Response:

We agree. We reconstructed the within-dataset baseline using group-aware splitting at the recording-unit level. Thus, all overlapping windows derived from a physical recording are assigned wholly to either the training or the test partition of a fold; no temporally adjacent overlapping windows are split between them. We also replaced the previous one-second windowing scheme with five-second windows and a 0.5-second overlap. The baseline results are being recomputed under this leakage-free design.

[TODO — not yet in the manuscript, and not yet recomputed.] The recording-unit grouping and the 5 s window with 0.5 s overlap exist in the code, but the Data preparation subsection of the manuscript still describes 1 s windows with a 0.125 s step — the very geometry the Reviewer objects to — and the leakage-free baseline has not been computed: the sweep is still running and no analysis has been produced from it. Reconcile the manuscript and confirm the numbers before this response is treated as final.

Comment 3

MR3. Several validation procedures are insufficiently defined. Dataset B trials appear to be treated as sessions without justification; the composition of the "task-only" datasets is unclear; class counts are absent; and cross-task testing does not separate task transfer from participant identity.

Response:

We agree and have corrected the terminology and validation design throughout.

Dataset B trials treated as sessions. The SAM40 repeated task recordings are treated as trials rather than as independent physical sessions. Cross-session validation is now restricted to the dataset with documented session entities, whereas SAM40 is evaluated with a separately named cross-trial protocol.

Composition of the "task-only" datasets. We now define every analytical subset explicitly by the conditions retained from the complete dataset mapping. The exact excluded conditions are stored in the resolved configuration and in the identity of the prepared data artifact, so that each task-specific result can be reconstructed unambiguously.

Missing class counts. We have added training and test class counts for every fold, together with per-class support in the classification tables. The revised Table 2 will report the relevant sample sizes alongside its performance measures.

Subject-identity confounding in cross-task testing. We rebuilt cross-task validation as leave-one-subject-out transfer: for each fold, the source-task training data exclude subject S, and the target-task test data contain only subject S. Source and target partitions are therefore subject-disjoint by construction.

[TODO — the class-count part is not true yet.] Fold composition and per-class support are computed by the analysis script, but the analysis has not been run and the manuscript contains no Table 2: the Results section is a cleared placeholder. The three other parts of this response — session versus trial terminology, explicit subset composition, subject-disjoint cross-task transfer — are implemented in the code and stated in the manuscript.

Comment 4

MR4. Reproducibility is incomplete within the manuscript. Filter type and order, ICA implementation and rejection criteria, STFT window and FFT parameters, feature dimensionality, scaler-fitting scope, XGBoost hyperparameters, tuning procedure, random seeds and fold construction are not adequately reported.

Response:

We expanded the reproducibility specification. The versioned configuration now records filtering, ICA method and EOG-proxy criterion, epoching, channel set, feature extraction, model hyperparameters, random seed, and validation protocol. Each run retains its resolved configuration. Hyperparameters are fixed a priori for the benchmark comparison; no data-driven hyperparameter tuning is performed, and this will be stated explicitly in the Methods.

[TODO — the configuration half is done, the manuscript half is not.] The versioned configuration does record filtering, ICA, epoching, channel set, features, model hyperparameters, seed and protocol, and every run stores its resolved configuration. The manuscript reports none of it: the Models subsection is a cleared placeholder, and the model family its text names (XGBoost) still has to be reconciled with the set of models actually configured — logistic regression, XGBoost, EEGNet, EEGConformer, ShallowNet and EEGPT. Write the Methods first, then finalise this response.

Comment 5

R1. The metric reporting is internally inconsistent. In single-label classification, support-weighted recall is mathematically equal to accuracy. Table 2 nevertheless reports, for example, 0.81 accuracy and 0.97 weighted recall for Full B, and 0.73 accuracy and 0.90 weighted recall for its cross-subject evaluation. Either the averaging description or the results are incorrect.

Response:

We agree and recompute all reported metrics from the saved window-level predictions rather than from independently maintained summary values. The revised analysis exports confusion matrices, per-class precision, recall and F1, balanced accuracy, macro-F1, MCC, AUROC where defined, and the training-fold majority-class baseline. The corrected Table 2 will replace the inconsistent earlier values.

[TODO — no corrected metrics exist yet.] Recomputation from saved window-level predictions is implemented in the analysis script, but it has not been run, and there is no Table 2 in the manuscript to correct. This response cannot be sent before the recomputed values exist and the inconsistency the Reviewer identified is demonstrably gone from them.

Comment 6

R2. Revise the presentation and interpretation. Figure 1 should include every validation family with uncertainty; Figure 2 should distinguish diagonal baselines from transfer cells.

Response:

We have redesigned Figure 1 to report every validation family and to show participant-level 95% confidence intervals for the displayed metric. The figure is generated directly from the recomputed results, rather than from a manually maintained plotting table.

We have also redesigned Figure 2 as a source–target transfer matrix. Matched within-composition baselines occupy the diagonal and are visually marked as such; all off-diagonal cells represent transfer. This separates the relevant within-condition reference from cross-condition performance.

[TODO — both figures are code, not manuscript.] The two designs described here are implemented in the analysis script, but neither figure has been generated, and the Results section of the manuscript currently contains no figures at all. The past tense is justified only once they are produced and included.

Comment 7

R3. The 15 references are relevant and include the original publications for both datasets, established cognitive-task literature and selected studies on EEG transfer. However, the coverage is too limited for the broad methodological claims advanced. Statements concerning DANN, ADDA, CORAL, TCA, Riemannian alignment, transformer architectures, self-supervised learning, XGBoost performance and low-density ICA are presented without appropriate citations.

Response:

We thank the reviewer for this comment, which prompted a full audit of the bibliography rather than a simple expansion of it. We have acted on it in two ways.

(a) Expanded framing and citations in the Introduction. The Introduction previously moved directly from the general difficulty of EEG generalization to the aims of the study, without grounding the central hypothesis. We have added two paragraphs. The first states why cross-paradigm transfer is expected to be fragile: transfer learning depends on domain *and label* compatibility between source and target, and attention lacks a single accepted operational definition, so datasets nominally labelled for "attention" may in fact probe different attentional types, producing label mismatch under domain shift. The second states the gap explicitly, adopts the term *zero-shot generalization* for what is measured here, and positions the study as a baseline against which transfer-learning and domain-adaptation methods can be compared. These paragraphs add five references that were absent from the submitted version (Wu et al. 2022; Zhang et al. 2020; Hommel et al. 2019; Sun et al. 2025; Li et al. 2025).

(b) Verification of the entire reference list. Prompted by the reviewer's remark on reference quality, every entry in the bibliography was resolved against Crossref, arXiv or the publisher record. This uncovered errors in the submitted reference list that we correct here and regret. Specifically: the cross-task workload study cited in Background was printed with an author list that does not belong to it (the correct authors are Ke et al., Front Neurosci 2021;15:703139); the single-channel Kalman/ELM study was printed with an incorrect author list, journal, year and DOI (the correct record is He et al., Front Hum Neurosci 2025;18:1481493); and three further cited entries could not be resolved to any existing publication and have been replaced by verified sources supporting the same statements (Esterman et al., Cereb Cortex 2013;23:2712–2723 for sustained attention; Kitaura et al., Clin Neurophysiol Pract 2017;2:193–200 for attentional demand during mental arithmetic; Tomasino and Gremese, Front Hum Neurosci 2016;9:693 for the mental-rotation network). Duplicate entries for the same work, and several entries with incorrect author names, volumes or DOIs, were also corrected. The revised reference list has been checked to contain no unresolved or duplicated records.

We note that part of this comment remains open. The passages discussing domain-adversarial and adversarial discriminative adaptation, correlation alignment, transfer component analysis, Riemannian alignment, transformer-based and self-supervised representation learning, gradient-boosting performance on EEG features, and the stability of ICA on low-density montages are still stated without supporting citations. We will supply appropriate references for each of these claims in the next revision round, together with the specific recent works requested by Reviewer 3.

[TODO — two uncited entries, and Background does not exist.] Part (a) is accurate: the two Introduction paragraphs and their five citations are in the manuscript. `verma2023attentionet` and `muhl2014workload`, however, are present in `refs.bib` but cited nowhere, so they do not enter the reference list; either cite them or drop them, and keep the count above in agreement with what is actually cited. The Background section — where the missing citations for domain adaptation, alignment, transformers, self-supervised learning, gradient boosting and low-density ICA belong — is a cleared placeholder, which is why the closing paragraph of this response is accurate today; rewrite it once that section exists.

Reviewer #2

Comment 1

Reconsider the target construct. The problem should be presented either as discrimination between operationally defined experimental conditions or as attention classification supported by independent attention measurements.

Response:

We have adopted the first of the two formulations the Reviewer permits, and the revised manuscript is built around it throughout.

The second formulation is unavailable for these data, and the manuscript now demonstrates this rather than assuming it. The authors of Dataset A report that no independent objective control could be introduced, because active probing of a disengaged participant destroys the state being measured and no external correlate of passive focus could be identified. The public release of Dataset B comprises raw and filtered recordings, artifact-removed signals, electrode coordinates and per-trial subjective stress ratings, but not the participants' responses, so performance-based labels cannot be constructed from it. Neither dataset can therefore support attention classification anchored to an independent measurement of attention.

[TODO — the manuscript does not yet carry this formulation.] What exists: the Methods state that the model was trained to predict, from the spectral features of a window, whether the participant was completing a high-demand or a low-demand condition during that window, and not to predict a level of attention. What does not exist: Table 1 still carries the classes "High Attention" and "Low Attention", no paragraph follows it stating what the labels do not denote, and the Abstract is a cleared placeholder, so it says nothing about the sense in which "attention decoding" is used in the title. The five label-mapping arms have no counterpart in the code either — see Comment 3.2. Rewrite this response together with the response to MR1, once the corresponding manuscript paragraphs are written.

We would add that this constraint is not peculiar to our two datasets. The same criticism has been published against vigilance and workload decoders whose labels encode the condition of the protocol rather than the state of the participant. The revision makes this the subject of the study rather than a caveat to it: since publicly available attention datasets supply protocol-defined labels, the question that can honestly be answered is how much of a decoder's performance survives when the operational definition of its labels is changed.

Comment 2

Sensitivity analyses should exclude drowsy or eyes-closed epochs and evaluate alternative label mappings.

Response:

We agree that drowsiness should not be silently merged with awake unfocused behaviour. The primary analysis now excludes the explicitly labelled drowsy segment, and alternative condition mappings are represented as separate, reproducible analyses. We cannot validly identify or exclude eyes-closed epochs from the available recordings: they contain no EOG channel, video, or epoch-level eye-state annotation. Frontal EEG channels were used only as proxies to identify EOG-like ICA components; they do not provide ground truth for eye closure. We therefore do not claim eyes-closed epoch rejection, because an unvalidated amplitude-based proxy could remove meaningful EEG activity.

[TODO — the alternative mappings do not exist.] There is no label-mapping sensitivity analysis anywhere: a run selects its conditions through `dataset.exclude_conditions`, and the sweep defined in the Makefile varies the validation protocol, the dataset and the model — never the mapping. No M0–M4 arms are defined in the manuscript or in the configuration, and none are being computed. Either implement the arms and describe them in the manuscript, or answer this comment with the narrower thing that is true — the drowsy exclusion above, and the per-task subsets already used for cross-task transfer — and say plainly that a full mapping-sensitivity analysis is outside the scope of this revision.

Comment 3

Reconstruct all validation splits at the participant, session, or trial level before window segmentation.

Response:

We reconstructed all validation protocols around recording-level metadata (participant, documented session, or trial) before assigning windows to folds. Every window from a recording unit remains on one side of a split. The revised Methods distinguish cross-session and cross-trial validation rather than using the terms interchangeably.

Comment 4

Every preprocessing transformation, including scaling and data-dependent feature processing, should be fit exclusively using the training data.

Response:

All learned transformations, including standardization, are contained in the estimator pipeline and are fit separately within each training fold. The handcrafted feature transforms are deterministic functions of an individual window and estimate no parameter from the data; they therefore have no fitting scope that could transmit information across folds.

Comment 5

Cross-task experiments should also use subject-disjoint source and target partitions.

Response:

We agree and rebuilt cross-task validation as leave-one-subject-out transfer. For each held-out participant, the source-task training partition excludes that participant and the target-task test partition contains only that participant. The source and target partitions are consequently subject-disjoint.

Comment 6

Audit and recompute all metrics from the saved predictions. Class counts, confusion matrices, per-class precision and recall, balanced accuracy, macro-F1, MCC, and AUROC should be reported alongside majority-class baselines.

Response:

All performance measures are now recomputed from saved window-level predictions. The revised outputs include fold composition, confusion matrices, per-class precision, recall and F1, balanced accuracy, macro-F1, MCC, AUROC where defined, and majority-class baselines. Table 2 will report these metrics and their definitions consistently.

[TODO — implemented, but neither computed nor reported.] Every measure listed here is implemented in the analysis script and none of it is invented; none has been produced yet, because the sweep is still running and the analysis has not been run over it. The manuscript has no Table 2 at present.

Comment 7

Fold- or participant-level distributions, confidence intervals, and appropriate paired statistical comparisons are also required.

Response:

The revised analysis reports fold-level and participant-level distributions and 95% confidence intervals with their aggregation unit stated explicitly. It also performs paired, participant-level Wilcoxon comparisons only when runs share the same target data, with Holm and false-discovery-rate adjustments for multiple comparisons.

[TODO — as for Comment 3.6.] Fold- and participant-level distributions, 95% confidence intervals and paired Wilcoxon tests with Holm and false-discovery-rate correction exist in the analysis script; they have not been run, and the manuscript reports none of them.

Comment 8

Provide a complete and reproducible description of the methods, including filter characteristics, the ICA procedure, STFT settings, band-power calculations, feature dimensions, hyperparameters, tuning strategy, and random seeds.

Response:

We have expanded the Methods and the versioned configuration to specify the filtering recipe, ICA method and EOG-proxy criterion, window geometry, channel selection, feature extraction, model hyperparameters, random seed, and fold construction. Hyperparameters are fixed before the benchmark comparison; no data-driven tuning is performed. Each run stores its resolved configuration so that the exact analysis can be reproduced.

[TODO — see the response to MR4.] The configuration side is done, the Methods side is not: the Models subsection of the manuscript is a cleared placeholder, and the model family named in its text contradicts the models actually configured.

Comment 9

At least simple linear and tree-based benchmark models should be included.

Response:

We added logistic regression as a linear benchmark and gradient-boosted decision trees (XGBoost) as a tree-based benchmark, evaluated under the same validation protocols as the deep models.

[TODO — confirm the final benchmark set, then report it.] The models configured today are logistic regression, XGBoost, EEGNet, EEGConformer, ShallowNet and EEGPT; the SVM and ExtraTrees named in earlier drafts of this letter are no longer in `configs/model/`. The Models subsection of the manuscript names none of them, and no results exist for them yet.

Comment 10

An alignment or domain-adaptation baseline would further strengthen the transfer analysis.

Response:

[TODO]

Comment 11

Revise the presentation and interpretation. Figure 1 should include every validation category with corresponding uncertainty estimates.

Response:

Figure 1 has been redesigned to include every validation family and participant-level 95% confidence intervals, generated directly from the recomputed results.

[TODO — see the response to Comment R2 of Reviewer 1.] The figure is implemented in the analysis script, has not been generated, and is not in the manuscript.

Comment 12

Figure 2 should distinguish diagonal baseline results from transfer cells.

Response:

Figure 2 has been redesigned as a source–target matrix in which matched within-composition baselines occupy and are explicitly marked on the diagonal; off-diagonal cells represent transfer.

[TODO — see the response to Comment R2 of Reviewer 1.] The figure is implemented in the analysis script, has not been generated, and is not in the manuscript.

Comment 13

Table 2 should include sample sizes and corrected metric definitions.

Response:

The revised Table 2 will include the number of participants, windows, and class support for each analysis, together with metrics recomputed from the saved predictions and clearly defined aggregation units.

[TODO — there is no Table 2 yet.] The Results section of the manuscript is a cleared placeholder; the sample sizes and metric definitions promised here can be written only after the analysis has been run.

Comment 14

Claims regarding hardware, causal language, references, and the secondary data-use ethics statement should also be corrected.

Response:

[TODO]

Reviewer #3

Comment 1

I have carefully reviewed the article. At a first glance, this paper would have been really very interesting but unfortunately it is really too short and superficial for the promises it makes at the beginning. The authors set up a compelling premise in the introduction, promising to systematically evaluate how neural markers of attention transfer across diverse paradigms and validation strategies. However, the execution falls completely flat. It would be necessary to compare all the most advanced deep models for EEG decoding at the state of the art and see if, by passing from one task to another, they do better or worse. Instead, here unfortunately there are very few discussions on state of the art models and on datasets used for many tasks such as motor imagery, sleep, and other cognitive states. The entire methodology relies on a simplistic feature extraction pipeline combined with a traditional machine learning classifier, which completely ignores the massive advancements made in deep representation learning for electroencephalography over the last several years.

Response:

We appreciate this concern. The revised benchmark evaluates classical feature-based models alongside EEG-specific deep architectures — EEGNet, EEGConformer, ShallowNet and EEGPT — which receive the prepared EEG windows directly rather than handcrafted spectral features and are evaluated under the same validation protocols. We have also narrowed the manuscript's claims to the two available datasets and their operationally defined conditions.

[TODO — the results and the state-of-the-art discussion are both missing.] The deep models are configured and are being run, but no results exist for them, and the Background section that would review deep EEG decoding is a cleared placeholder. This response cannot stand as written until both are in place.

Comment 2

When discussing the state of the art in electroencephalography decoding, it is entirely unacceptable to limit the analysis to gradient boosted decision trees like XGBoost fed with basic short time Fourier transform spectral power features. The field has moved well beyond these classical baselines. Advanced deep models such as convolutional neural networks specifically designed for time series, spatial temporal graph convolutional networks, and transformer based architectures have become the standard for brain computer interfaces and cognitive state monitoring. The manuscript barely touches upon these architectures, briefly mentioning transformers in the related work but failing to implement or benchmark any of them. To truly understand whether neural signatures of attention are invariant across paradigms, one must leverage models capable of learning hierarchical spatial and temporal representations directly from the raw or minimally preprocessed signals. By relying on predefined spectral bands and overlapping windows, the authors impose a rigid inductive bias that likely destroys the subtle, task invariant dynamics they claim to be looking for. Therefore, the conclusion that current classifiers capture context dependent features rather than invariant signatures is heavily confounded by the fact that the authors used a very rudimentary classifier that lacks the capacity to learn invariant representations in the first place.

Response:

We added EEG-specific deep architectures — EEGNet, EEGConformer, ShallowNet and EEGPT — as end-to-end benchmarks on the prepared raw EEG windows, alongside the feature-based logistic regression and XGBoost baselines. This comparison tests whether the observed validation-pattern differences persist when models can learn representations directly from the signal rather than only from predefined spectral features.

[TODO — as for Comment 4.1, no results exist yet.] Earlier drafts of this letter also named SVM and ExtraTrees, which are no longer configured; keep the list here in agreement with `configs/model/` and with the Models subsection of the manuscript once that subsection is written.

Comment 3

Furthermore, the narrow focus on a passive control task and short cognitive tests like the Stroop test and mental arithmetic severely limits the scope of the claims regarding cross task generalization. A comprehensive evaluation should have included datasets spanning a much wider variety of neurological tasks. For instance, motor imagery is one of the most widely studied paradigms in the field, and analyzing how attentional engagement during motor imagery transfers to active cognitive tests or sleep staging would have provided a much stronger foundation for the paper. Sleep datasets, in particular, offer a continuous spectrum of arousal and attentional states that could have perfectly complemented the drowsy condition in the first dataset. The lack of discussion on these broader applications and the datasets commonly used for them makes the paper feel isolated from the broader brain computer interface community and leaves the reader wondering how these models would perform in real world clinical scenarios.

Response:

We do not extend the evaluation to motor imagery or sleep staging, and we want to be explicit about why this is a scope boundary rather than a limitation we intend to resolve with more data. Motor imagery and sleep staging are defined by different operational constructs than the sustained-attention and cognitive-load states this paper targets: motor imagery decoding identifies sensorimotor-rhythm modulation tied to imagined movement, and sleep staging assigns polysomnography-defined vigilance stages. Neither is a measure of attentional engagement during an ongoing task. Adding them would not test whether attention-decoding models transfer across attention-related paradigms; it would test transfer across constructs the models were never designed to represent, which is a different research question from the one this paper asks. We therefore keep cross-task and cross-dataset evaluation within operationally defined attention-related paradigms and state this as an intentional scope boundary in the manuscript, rather than implying that a wider survey of unrelated BCI paradigms would have strengthened the present claims.

[TODO — this scope boundary is nowhere in the manuscript.] The argument stands on its own and needs no new data, but no section currently makes it: the Background section, where it belongs together with the discussion of the motor-imagery and sleep paradigms the Reviewer asks about, is a cleared placeholder.

Comment 4

To rectify these significant omissions, the authors must update their literature review and theoretical framework by citing and discussing recent works that successfully apply deep learning to complex electroencephalography tasks. I strongly require the authors to cite the paper titled *Investigating the impact of rational dilated wavelet transform on motor imagery EEG decoding with deep learning models* published in the year 2025. It is extremely important to cite this work because it clearly demonstrates how advanced time frequency transforms can be integrated with deep learning models to significantly improve decoding performance on complex tasks like motor imagery. This highlights the inadequacy of the standard short time Fourier transform approach used in the current manuscript. Additionally, the authors must cite the paper titled *RatioWaveNet: A Learnable RDWT Front-End for Robust and Interpretable EEG Motor-Imagery Classification*, also published in 2025. Citing this paper is crucial because it showcases the current state of the art in developing robust, interpretable front ends that are completely learnable, effectively replacing the static feature extraction pipelines that the authors currently rely on. Since these papers are very recent, they represent the cutting edge of the field and provide exactly the kind of state of the art methodological context that this superficial manuscript currently lacks.

Response:

[TODO]

Comment 5

Moving to the semantic and conceptual issues within the manuscript, there are several major flaws that undermine the validity of the unified labeling scheme. The authors decided to merge the drowsy state, the unfocused state, and the resting state into a single low attention class. This is a severe semantic error. From a neurophysiological perspective, drowsiness involves a transition into early stage sleep, characterized by the emergence of theta waves, vertex sharp waves, and a general withdrawal of environmental awareness. This is fundamentally different from an awake but unfocused state, which might involve default mode network activation and mind wandering, or a resting state with eyes open or closed, which is typically dominated by posterior alpha oscillations. By grouping these distinct physiological phenomena into a single binary class, the authors create an artificially heterogeneous category that confuses the classifier. This semantic misalignment invalidates many of the cross dataset conclusions, as the model is likely learning to distinguish arousal levels rather than attention levels.

Response:

We accept this criticism and have acted on it rather than argued against it. We agree that drowsiness is not interchangeable with an awake unfocused state or with eyes-open rest, and that the neurophysiological grounds the Reviewer gives — the transition towards early-stage sleep, with its own signature and with sustained eye closure — are sufficient reason not to place these conditions in one class.

Drowsy epochs are therefore excluded from the primary analysis entirely, and not reassigned or flagged as a limitation. The exclusion is implemented as a step of the data-preparation pipeline, ahead of channel selection, filtering, artifact removal and windowing, so that drowsy data do not enter the ICA decomposition or the fitting of the feature scaler either. The low-demand class of the primary mapping now contains only the unfocused and resting conditions, and the manuscript states the Reviewer's own reasoning as the justification for the exclusion.

[TODO — the remainder of this response describes a manuscript that does not exist.] Table 1 has not been renamed: it is still "Unified binary attention label mapping" with the classes "High Attention" and "Low Attention", and no paragraph follows it. There is no sensitivity arm M0, in the manuscript or in the code, so the Reviewer's inference about arousal is still not testable. There is no Limitations subsection at all, although two passages of Methodology already refer the reader to one. Write the Table 1 reframe and the Limitations subsection first; only then can this response state what the arousal objection is actually answered with.

Comment 6

Furthermore, framing mental arithmetic purely as an attention task ignores the massive working memory load it requires, which produces entirely different cortical activation patterns compared to the simple visual monitoring required in the train simulator task.

Response:

We accept the observation and have not attempted to argue it away. Under the operational definition adopted in this revision, the working-memory load of mental arithmetic is not a confound to be removed from an attention contrast but a constituent of the high-demand condition, and mental arithmetic is no longer framed as a pure attention task.

We also draw the broader consequence that follows from the Reviewer's point. The attention-decoding and mental-workload-decoding literatures operationalize their targets in the same way — task versus rest, or high versus low demand — and are to that extent operationally indistinguishable. This is a further instance of the main argument of the revision rather than an exception to it: where the label is supplied by the protocol condition, the boundary between the two constructs is not drawn by the data.

[TODO — three claims were removed from this response, each unsupported.] (i) That cross-task transfers involving arithmetic were consistently the weakest: those numbers come from the superseded pipeline, were removed from the manuscript, and have not been recomputed. (ii) Label-mapping arm M4, which would keep the Dataset B tasks separate: no such arm exists in the configuration or in the sweep, and nothing of the kind is being computed. (iii) The alpha-desynchronization and frontal-midline-theta rationale for aligning the two protocols: the manuscript still argues that alignment from the psychological labels of the individual tasks, which is exactly what the Reviewers did not accept. Rewrite this response once the manuscript carries a mechanism-based rationale and once cross-task numbers exist.

Comment 7

Similarly, the grammatical and syntactic quality of the manuscript leaves much to be desired. There are numerous instances of poor phrasing and missing punctuation that disrupt the reading flow and obscure the scientific meaning. For example, in the description of Dataset A, the authors write that all experiments were performed with volunteer participants chosen among students. This phrasing is awkward and unscientific; it would be much better to state that the participant cohort consisted of university student volunteers. In the description of Dataset B, there are glaring punctuation errors in the descriptions of the tasks. The text reads Stroop Color Word Test participants were required to identify the color, and Arithmetic problem solving participants mentally solved a mathematical problem, and Mirror Image Recognition participants were shown images. In all these cases, there is a missing colon or em dash after the name of the task, making it seem like the participants themselves are named after the tasks. These are careless grammatical errors that should have been caught during proofreading. Furthermore, the phrasing in the results section, specifically the sentence stating that accuracy and F1 score reaching 0.86 for Dataset A and 0.81 and 0.88 respectively for Dataset B, is disjointed and confusingly structured. The authors also failed to properly format the cross dataset directions in their text, sometimes omitting the necessary words to explain the direction of the transfer.

Response:

[TODO]

Comment 8

Another critical methodological failure that must be addressed is the cross task validation experimental design. The authors openly admit in the limitations section that in their cross task experiments, the source and target tasks were drawn from the same pool of Dataset B participants, and consequently, task transfer and subject identity are not fully disentangled. This is not just a minor limitation; it is a fatal flaw for a study whose entire premise is about evaluating transferability and generalization. If the model is evaluated on a new task but with the same subjects it was trained on, any successful transfer could merely be the result of the classifier learning subject specific noise profiles, baseline anatomical features, or electrode impedance artifacts, rather than genuine task invariant attention markers. A true cross task evaluation must be performed using a leave one subject out approach across tasks, ensuring that the model is tested on both unseen tasks and unseen subjects simultaneously, or at the very least, evaluated on completely disjoint subsets of subjects.

Response:

We agree and rebuilt the cross-task protocol as leave-one-subject-out transfer. In each fold, source-task training data exclude the held-out participant and target-task testing uses only that participant. The revised design therefore tests both an unseen task composition and an unseen participant.

Comment 9

The authors also acknowledge that their within dataset baselines are likely overly optimistic due to the use of densely overlapping windows. They state that overlapping one second windows with a zero point one two five second step size lead to highly autocorrelated samples, and that splitting these windows randomly across cross validation folds causes temporal leakage. While it is good that they acknowledge this, merely stating it in the limitations is insufficient for a paper focused on validation strategies. The core contribution of this paper is supposed to be a rigorous evaluation of validation schemes, yet they purposefully include a flawed baseline that they know is leaking data. They should have implemented the group aware splitting by trial or session that they suggest as future work directly in this manuscript. Presenting inflated within dataset metrics undermines the entire comparative analysis, as the drop in performance observed in cross subject and cross session scenarios might just be the result of removing data leakage rather than actual domain shift. The failure to correct this known issue makes the baseline metrics virtually meaningless.

Response:

We agree and removed this leakage from the baseline. Overlapping windows are grouped by their recording unit, and group-aware cross-validation assigns every window from a unit to either training or testing, never both. The revised baseline results are recomputed with this procedure.

[TODO — the manuscript still describes the old windows.] Group-aware splitting is implemented, but the Data preparation subsection still states 1 s windows with a 0.125 s step, which is the geometry the Reviewer objects to; the configuration now uses 5 s windows with 0.5 s overlap. The recomputed baseline does not exist yet.

Comment 10

Additionally, the reliance on a reduced twelve channel montage significantly limits the potential of the study. While aligning the spatial resolution of the two datasets is a logical step for cross dataset evaluation, dropping from thirty two channels down to twelve discards a massive amount of spatial information that could be vital for deep learning models to learn invariant representations. The authors note that this low density configuration limits the stability of the independent component analysis used for artifact removal, which introduces yet another confounding variable. If the artifacts were not cleanly removed due to the lack of spatial channels, the classifier might simply be tracking eye movements, blinks, or muscle tension rather than cortical attention networks. This is especially problematic given that the Stroop test and mental arithmetic often elicit different physical responses, facial microexpressions, and stress induced muscle artifacts compared to a passive train driving simulator. The failure to adequately separate these physiological artifacts from true neural signals casts doubt on whether the model is actually decoding attention at all.

Response:

We clarify that the shared 12-channel montage is selected only after artifact correction. For each recording unit, ICA is fitted to, and applied on, all EEG channels available in that original recording; channel selection occurs only after ICA application and before epoching and model fitting. The common-channel restriction therefore does not reduce the dimensionality of the ICA decomposition. It is used only to give the cross-dataset models an identical set of input channels.

We nevertheless acknowledge an important residual limitation. Neither source dataset provides an independent EOG channel, so ocular-component candidates are identified using pre-specified frontal EEG channels as EOG proxies. This permits consistent automated component selection, but it is not ground-truth validation of ocular-artifact removal. We have made this limitation explicit and have restricted our interpretation accordingly: the analyses compare classification of operationally defined experimental conditions and do not establish that the models isolate cortical attention signals from all possible physiological artifacts.

[TODO — the manuscript contradicts this response.] The code does what is described here: ICA is fitted and applied on all channels of the original recording, and the common 12-channel montage is picked afterwards (`src/preparation.py`). The Data preparation subsection of the manuscript states the opposite order — channel selection first, ICA afterwards. Fix the manuscript before this response is sent.

Comment 11

Finally, the absence of any discussion regarding model explainability or feature importance is a missed opportunity. Even with a simpler model like XGBoost, the authors could have analyzed which spectral bands and channels were most critical for the cross task and cross dataset predictions. Without an analysis of the underlying features driving the decisions, the claim that models capture context dependent features remains purely speculative. State of the art deep models, such as the ones recommended for citation earlier, often include interpretability modules or attention mechanisms that allow researchers to visualize the spatial and temporal features the network relies on. By neglecting this aspect, the authors fail to provide any neuroscientific insights into why the transfers failed, leaving the paper as a purely empirical exercise in applied machine learning with poor experimental controls.

Response:

We added fold-wise feature attribution for the models that expose it directly: coefficients for logistic regression and gain-based importances for XGBoost. The analysis maps these values back to the configured channel order and frequency bands and reports their fold average. We present this as model-specific descriptive evidence, not as an explanation of the deep models.

[TODO — saved, but not yet aggregated or reported.] Per-fold attributions are stored with every run, but the analysis that aggregates them has not been run and the manuscript has no section reporting them. Earlier drafts of this letter named ExtraTrees, which is no longer configured.

Comment 12

In conclusion, the manuscript fails to deliver on its ambitious initial premises. To be considered for publication, it requires a complete overhaul. The authors must replace their obsolete feature extraction and gradient boosting pipeline with a suite of advanced deep learning architectures, comparing their performance across various transfer scenarios. They must expand their dataset inclusion to encompass a wider variety of cognitive states and tasks, particularly motor imagery and sleep stages, to provide a true measure of paradigm invariance. They must correct the severe semantic errors in their label harmonization process, particularly the grouping of sleep related states with resting states, and fix all grammatical mistakes and punctuation omissions throughout the text. Most importantly, they must implement rigorous, leakage free validation schemes for all their baselines and ensure that cross task evaluations are completely separated from subject identity confounds. Without these major revisions and a significant expansion in scope, the paper remains too superficial and methodologically flawed to contribute meaningfully to the state of the art in electroencephalography decoding.

Response:

We appreciate the detailed assessment. The revision replaces the leaking baseline and the subject-confounded cross-task protocol, recomputes metrics from saved predictions, adds participant-level uncertainty and paired comparisons, and benchmarks linear and tree-based models against EEG-specific deep architectures (EEGNet, EEGConformer, ShallowNet, EEGPT). We also exclude the explicitly labelled drowsy segment from the primary analysis and narrow the interpretation to operationally defined attention-related conditions, deliberately outside the scope of unrelated BCI constructs such as motor imagery or sleep staging. The remaining detailed responses specify the corresponding changes and limitations.

[TODO — summary response, to be finalised last.] It can only be written once the individual responses above are settled: the sweep is still running, no analysis has been produced from it, and the Abstract, Background, Models and Results sections of the manuscript are cleared placeholders.